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Optimizing ESP Well Recovery: A Comparative Case Study on Effective Chemical Solutions for Schmoo Deposit Removal

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Summary

Oil fields with increasing water often suffer from schmoo deposits in their wells and processing facilities, containing mixtures of oil with inorganic scale and corrosion products. Schmoo deposits are oil-wet and sticky and cause multiple problems in oil and gas production, like production impairment from formation damage, poor ESP performance, plugging of tubulars and malfunctioning of multi-phase flow meters. Cleaning of schmoo-affected wells by chemical solvents is possible, although several studies have indicated that multi-step treatments with organic solvent followed by acid provide better results than single solvent treatments. However, the ultimate effectiveness of the treatment depends on the composition of the schmoo deposits. Therefore, to design the optimum mitigation strategy it is critical to understand the exact nature of the deposits and underlying root causes. This paper focuses on severe schmoo deposits in an onshore carbonate oilfield in the UAE, producing from one reservoir that consists of three separate flow units. The presence of fractures as well as high permeability dolomite streaks result in severely non-uniform inflow along the horizontal section of oil producer wells. This results in accelerated water break-through and requires artificial lifting by Electrical Submersible Pump (ESP) in many oil producing wells. Multiple ESP wells suffer from re-occurring schmoo deposits, downhole as well as in the ESP string and multiphase flow meters. Deposit analysis showed the presence of oil, carbonate and sulfate scale, iron sulfide (FeS) and silica. Chemicals capable of dissolving both organic and inorganic fractions were selected in collaboration with the stimulation contractors. The chemicals were also tested for thermal stability and compatibility with ESP elastomers. The chemicals were placed by bullheading (BH) and/or coiled tubing (CT). Production was restored after a variable soaking period. A total of seven chemical strategies were tested, with different combinations of (mutual) solvents and dispersants, applied alone or followed by hydrochloric acid. The success of the chemical treatment was analysed and expressed as percentage gain of ESP intake pressure and production rate, by comparing the values post-treatment to the values prior to treatment. Overall, the chemical soaks yielded significant results. The average production gain for chemical treatment by CT or BH was very similar. Cleaning jobs executed in two steps (solvent followed by acid) were more successful than single-step jobs (solvent or dispersant only). The success rate of dual-step jobs was 81%. The average production gain of dual-step jobs was 140%, compared to 70% for single-step jobs. When the dual-step

approach was applied there was no big difference between the different organic phases (solvent, dispersant or mutual solvent). This study for a Middle Eastern carbonate field shows that a standardized two-step chemical treatment using organic solvents followed by hydrochloric acid effectively removed schmoo and restored well production while maintaining equipment integrity. The approach proved successful in dissolving deposits and enhancing ESP performance and production rates.

Introduction

Many ageing fields with increasing water cut suffer from mixed deposits in their wells and processing facilities. These mixed deposits are often referred to in the oil and gas industry as *schmoo* and described by Littman (1987) as 'gelatinous mass tightly adhered to the metal surface containing large amounts of oil and other components like iron sulfate and calcium sulfate'. Although there is not a universal schmoo composition, all publications refer to a mixture of organic and inorganic compounds (Bohon *et al.*, 1998, Horsup *et al.*, 2007, Jones *et al*., 2010, Eroni *et al*., 2015, Lu *et al.*, 2024). Eroni et al. (2015) systematically analysed several mixed deposits and presented an elegant classification matrix that explains which types of schmoo are common in different part of injection and production systems.

One of the constituents that is generally always present is iron, mostly associated with iron sulfide, iron carbonate or iron oxides. Chen et al. (2019) evaluated possible mechanisms of FeS formation and concluded that in sour gas reservoirs it is unlikely that ferrous iron Fe^{2+} comes from the formation but rather from corrosion during production or stimulation. Other publications connect the presence of iron sulfide in schmoo to bacterial activity (Littman, 1987, Reeves, 1989, Hsi *et al.*, 1994).

Schmoo deposits are oil-wet and sticky and cause multiple problems in oil and gas production, like production impairment from formation damage, poor ESP performance, plugging of tubulars and malfunctioning of multi-phase flow meters. Cleaning of wells affected by schmoo by chemicals is possible, although a single solvent might be challenging as it will not always dissolve all constituents. For example, a set of different chemical solutions were applied to mixed deposits from water injection systems by Jordan et al. (2022). They demonstrated that batch treatment can be successful, continuous injection was partially successful as there was a reduction in the FeS content but not in sulfate scale. Alharith et al. (2021) dissolved mixed deposits by thermochemical reaction during lab testing, although no specific details were provided on the nature of the deposit. Treatment of mixed deposits with biocide by Hsi et al. (1994) showed increased injectivity in core plugs, indicating that injectivity loss is partially caused by bacteria. Lab studies by Lu et al. (2024) showed that cleaning schmoo deposits with organic solvent before dissolving FeS with acid makes a big difference. Their study indicated that acid was effective for several of the common iron sulfide minerals, and that acid strength did not make a large difference. These studies highlight it is critical to understand the exact nature of the schmoo deposits and underlying root cause to design the optimum mitigation strategy.

This study focuses on severe schmoo deposits in an onshore field in the UAE. The field is located south of Abu Dhabi and was discovered in 1966. Production started in 1983. The field is producing from one carbonate reservoir, that consists of three separate flow units. The reservoir is under water drive mechanism and the aquifer has been identified to have moderate support to the reservoir pressure. The reservoir pressure has dropped from the original pressure of 1970 psi to approximately 1450 psi in 2020, but has dropped further to below 1000 psia in certain sections. The main reservoir challenges are the water cut increase and pressure depletion. The reservoir fluid has an average bubble point pressure of 422 psia, solution gas to oil ratio (GOR) of 79.4 scf/stb, oil density of 0.815 g/cc and oil viscosity of 2.3 cP at bubble point pressure. The fluid composition analysis indicates that the content of H_2S is 0.02 mole% and CO_2 is 0.19 mole%. The field started water injection for reservoir pressure maintenance in 2019. The reservoir is very heterogeneous, cores and image logs showing the presence of fractures in many wells, as well as the presence of high permeability dolomites. These variations in permeability result in severely non-uniform inflow along the horizontal section of oil producer wells and accelerate the water breakthrough. The high water cut required artificial lifting by ESP in many oil producing wells.

The low permeability of the oil-bearing sections and heterogeneity are impacting productivity of some wells and impacts ESP performance (low pump intake pressure, frequent ESP tripping). Multiple wells suffer from partial or complete plugging due to formation of deposits in the well. Deposits have been observed in the horizontal 6" open hole section, in the ESP pump and string as well as in surface facilities like the multi-phase flow meter.

In this paper we will provide the results of a comprehensive study to identify the nature of the deposits and the outcome of multiple field trials to clean the wells and mitigate ESP performance. An overview will be presented of the different deposit types obtained from the field, together with an explanation of the root-cause of these deposits. An extensive chemical cleaning program was initiated, by using different chemicals and by applying different placement strategies. The ESP elastomeric internals were not compatible with some of the solvents, which required an alternative approach for some wells. The different chemical cleaning jobs will be presented with an after-the-job analysis of the success rate. The paper will finish with a summary of the study and recommendations for future applications.

Materals & Methods

Well overview

Table 1 below gives an overview of the wells that are part of the chemical treatment program outlined in this paper. All these wells produce from the same reservoir. These are the wells where a significant drop in ESP intake pressure was recorded, or a significant drop in production losses. These declines are determined by comparing the pressures and production rates to the ones from the moment the ESP wells were commissioned. Some wells are listed multiple times, because the problems re-occurred after treatment.

Table 1—Overview of the wells with recorded ESP problems, and part of the chemical treatment pilot study. The table shows if the ESP strings are equipped with a Y-tool, which dictates the chemical placement strategy.

Well	Date	Y-tool	WC	P decline [%]	Q decline [%]
1	1-Nov-24	N	24.4	0	93
2	8-Jun-25	N	70.7	15	84
3	5-Apr-24	Y	61.5		85
4	26-Mar-24	Y	29.8		78
5	21-May-25	N	67.9	14	0
6	6-Jan-25	Y	56.5		5
7	2-Mar-24	N	33.1	17	10
8	7-Sep-24	Y	23.8	0	0
9	3-Nov-24	N	34.5	23	0
10	26-May-24	N	41.8		88
11	7-May-24	Y	22.4	41	23
12	2-Jul-23	N	35.9	59	43
12	10-Jun-25	N	34.7	57	76
13	21-Dec-23	Y	23.1		53
13	4-Apr-25	Y	46.1		70
14	28-Jun-24	Y	20.8		73
14	19-Apr-25	Y	40.9	37	57
15	20-Mar-22	N	27.7	23	43
16	2-Apr-25	Y	22.7	48	56
17	1-Nov-23	N	26.7	40	78
17	4-Aug-24	N	25.0	56	99
17	18-Jan-25	N	31.6	0	93
18	25-Mar-24	N	18.4	24	32
18	3-Nov-24	Y	23.0	23	78
19	7-Jun-24	Y	12.5	6	44
20	5-Jul-24	Y	1.5	0	70
21	6-Jan-24	Y	3.4	37	12
22	10-May-25	N	16.7	0	94
23	7-Sep-24	Y	1.1	15	51
24	22-Mar-24	Y	3.5	39	68
24	01-Sep-24	Y	2.6	56	5
25	12-Dec-23	Y	37.5	13	60
26	2-May-25	Y	40.4		25
27	12-Jun-24	Y	21.4	43	72
28	13-Mar-25	Y	22.7	0	0

Water analysis

Table 2—Detailed water composition. Produced water is the combined water from flow unit 1 and 2. The formation water is the composition of brine from flow unit 1 only. Aquifer water comes from the shallow aquifer and is injected for reservoir pressure maintenance.

	Produced Water	Formation Water	Aquifer Water
Collected	Storage tank	Wellhead	Wellhead
Sodium	56200	54300	48200
Potassium	925	3300	1100
Calcium	14200	11300	9420
Magnesium	2300	1810	3050
Barium	1.2	1.8	0.25
Strontium	805	820	585
Dissolved Iron	0.01	0.73	0.25
Chloride	122940	110760	102860
Bromide		765	1222
Sulfate	280	315	630
Bicarbonate	145	285	240
TSS	10	65	244
TDS	225800	192187.5	227000
SG	1.138	1.128	1.116
Resistivity	0.062	0.064	0.069
pH @ 25C	6.31	6.8	7.79

Oil analysis

Table 3—Reservoir fluid composition of flow unit 1.

	Flow unit 1		Flow unit 1
N2	0.191	N2	0.13
CO2	0.059	CO2	0.15
H2S	0	H2S	0.08
C1	4.47	C1	6.74
C2	1.37	C2	1.57
C3	2.735	C3	3.59
iC4	1.306	iC4	1.34
nC4	2.736	nC4	3.11
iC5	1.94	iC5	1.97
nC5	2.077	nC5	2.19
C6	5.221	C6	3.56
C7	5.724	C7	5.91
C8	6.074	C8	6.39
C9	4.759	C9	5.8
C10	4.795	C10+	57.47
C11	4.063		
C12+	52.48		
C12+ MW	359.4	C10+ MW	292.26
C12+ density	0.9106	C10+ density	0.874
GOR [scf/bbl]	44	GOR [scf/bbl]	78
API	30.4	API	30.5
P-RES [psi]	1446	P-RES [psi]	1550
T-RES [F]	170	T-RES [F]	170
P-SAT [psi]	215	P-SAT [psi]	298

Oil SARA & WAT

Table 4—SARA composition of flow unit 1 and 2.

	Saturates	Aromatics	Resins	Asphaltenes
Flow unit 1	35.2	49.0	12.7	3.1
Flow unit 2	57.9	24.9	14.1	1.7

Table 5—Wax Appearance Temperature (WAT) for reservoir 1

	Reservoir 1
8000 psi	102°F
5000 psi	96°F – 111°F
500 psi	99.1°F
15 psi	98.8°F – 100.4°F

Field deposit analysis

Table 6—High level breakdown of field deposits. The well numbers correspond to the wells listed in Table 1.

	MPFM 1	MPFM 2	Well 4	Well 4	Well 12	Well 21	Well 21	Well 21	Well 23	Well 23
water soluble	9	6.5							5.7	3.7
organic	71	42	35-50	40	71	9		50-70	62	55
acid soluble	12	43	42	62		78	95	27	15	23
acid insoluble	0.1	0.2							8.7	12.5
Comments (*)	2, 5	2, 5	4	3, 5	2	3, 4, 6		3	1	1

(*) Comments:

1. Contains silica
2. Contains iron
3. Contains FeS
4. High solubility in mutual solvent and 15% HCl
5. Contains asphaltenes
6. Contains $CaCO_3$

Selection of wellbore cleaning chemicals

Traditionally, wells are cleaned by 15% hydrochloric acid in cases of inorganic scale and by Heavy Aromatic Naphtha (HAN) in case of organic deposits. HAN is generally a good solvent for the type of deposits encountered in the UAE Onshore wells, and cost-competitive to other organic solvents. However, it is clear that the deposits in this field are mixed deposits. Therefore, just 15% HCl or just naphtha might not give the best results.

Table 7—Chemical treatment strategy

Method	Company	Execution	Product
1	1	Single	Dispersant
2	1	Dual	(1) Dispersant, (2) acid
3	2	Single	Organic solvent
4	2	Dual	(1) Organic solvent, (2) acid
5	3	Single	Mutual solvent
6	4	Single	Naphtha
7	4	Dual	(1) Naphtha, (2) Acid

For wells where bullheading is the only option the chemicals are pumped through the ESP string. However, the ESP internals and cable splice contain elastomeric material that is not compatible with some organic solvents, naphtha in particular. The ESPs used in the field of this paper contain Aflas and EPDM. Table 8 below shows the results of chemical compatibility testing conducted by two chemical suppliers. Although one supplier indicated that aromatic solvents can be used with Aflas, another chemical supplier confirmed that the Heavy Aromatic Naphtha (HAN) used in our wells is not compatible with both Aflas and with EPDM.

Table 8—Chemical compatibility of different organic solvents with elastomers present in the ESPs used in our wells.

	Supplier 1		Supplier 2
	Aliphatic solvent	Aromatic solvent	HAN
Aflas (FEPM)	OK	OK	Not compatible
EPDM (CL-80)			Not compatible
NBR			Not compatible
HNBR	Up to 275°F	Up to 250°F	

Results

Deposits from 10 locations were analysed, and a rough breakdown of their main constituents was given in table 6. The deposits were collected from downhole, ESP or surface facilities like Multi-Phase Flow Meter. Some deposits were analysed in an external lab with full breakdown of the organic and inorganic fraction by applying solvent washing, acid washing and combustion (Loss on Ignition, LOI) at 500C and 900C. The liquid extracts from the washing steps were then analysed for its main ions by ion chromatography and Inductive Coupled Plasma (ICP). Some other deposits were given to contractor labs to identify solvents for effective dissolution, and only a high-level indication of the nature of the deposits was retrieved from these solvent selections. All deposits were mixed deposits, with an organic fraction and inorganic fraction. The inorganic fraction was further split into an acid-soluble and acid-insoluble fraction. From the analysis a rough average breakdown was determined, presented in figure 1.

For the selection of effective solvents for the field deposits it is critical to understand the nature of the different fractions. For the inorganic fraction scale simulations were used to check what minerals are likely to form under the field conditions. For the organic fraction some results were used from a PVT and flow assurance study of recently collected bottom hole samples from a newly drilled well.

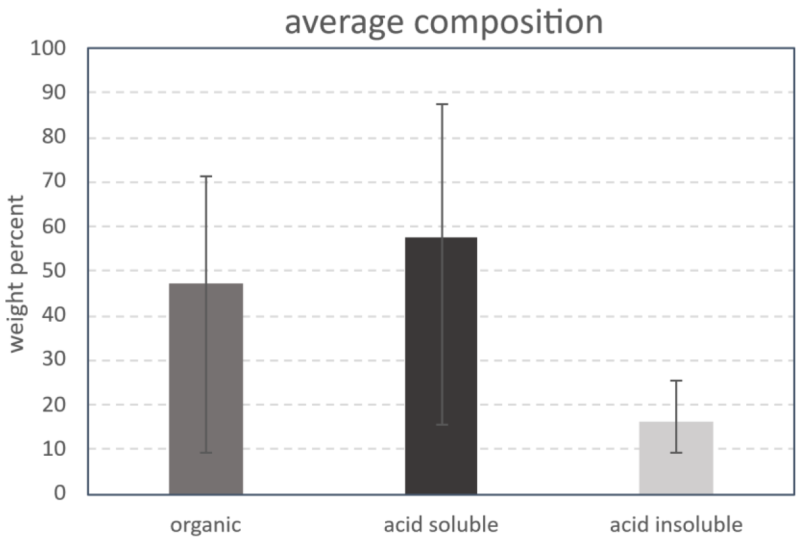

Figure 1—Breakdown of field deposit key constituents, based on the analysis shown in Table 5. The bars show the average composition, and the black lines indicate the highest and lowest recorded value. Since only two field deposits had recorded water-soluble components, this was excluded from this graph.

The oil and hydrocarbon compositions were used for inorganic scale simulations, by OLI Studio, version 11.0.1.9. For the scale simulations two different scenarios were assessed; (a) self-scaling from the production of formation water only, and (b) injection water (used for reservoir pressure maintenance) breakthrough in the oil producers. The simulations focus on the formation of common types of carbonate scale, sulfate scale and sulfide scale, from the reservoir to the surface. Table 9 below shows the different simulation nodes, i.e. pressure and temperature combinations.

Table 9—Pressure and temperature combinations, used in OLI Studio 11.0.1.9 for inorganic scale calculations.

Simulation node	T [F]	P [psi]
Reservoir - initial	170	1550
Reservoir - current	170	600
Downhole	170	400
Downhole - low	170	200
ESP in	150	200
ESP out	290	800
Wellhead	110	600
Surface	90	15

The formation water composition, as shown in table 2, was measured in an external laboratory. However, dissolved gases CO_2 and H_2S are measured by stain tubes in the field and recorded in separate files. The concentration of dissolved gases were added to the formation water composition. The formation water was then equilibrated with the limestone formation to ensure it was saturated with calcium carbonate at reservoir conditions. The scaling risk was calculated for a water cut of 50%. For the breakthrough of injection water the same approach was followed. The concentrations of dissolved gases, from stain tube measurements in the field, were added to the injection water composition. The injection water was then equilibrated with the reservoir to calcium carbonate saturation. The simulations were done with an overall water cut of 50%, with an injection water fraction of 0.2. The water compositions listed in table 2, based on recently collected water samples, show a low concentration of Fe^{2+}. Historic water data had shown an iron concentration up to 47 mg/l. It is unknown if this concentration of Fe^{2+} is the result of corrosion or naturally present in the formation. Since certain sections of the reservoir contain H_2S it is not likely that the concentration in the formation brine is that high, as it would have been scavenged by H_2S to form FeS. However, an extra sensitivity was included with an Fe^{2+} concentration of 10 mg/l to evaluate if that changes the risk for scale formation.

The simulation results are shown as Scaling Tendency (ST), sometimes referred to as Saturation Ratio (SR). The ST is determined by the concentration of scaling ions divided by the solubility product of the mineral formed by these ions. When the ST > 1 it means that the mineral is oversaturated in the brine and could precipitate as a solid. However, the ST is purely a thermodynamic value, in reality the ST can reach higher values before it really becomes problematic. Although the precise ST value above which precipitation occurs depends on various field factors, it is often assumed that $CaCO_3$ and $SrSO_4$ does not occur below ST values of 3 to 5. Iron sulfide FeS is a very poorly soluble mineral, and precipitation will probably occur at lower ST values between 1 to 3.

The simulation results are displayed in Figure 2, for the minerals that are oversaturated and have a risk for scale precipitation and deposition. The figure shows the simulation results for the current reservoir pressure of 600 psi and an assumed bottomhole pressure of 200 psi. This low bottomhole pressure was selected because there are indications that the wells with low bottomhole pressure (often close to or below bubble point) suffer from the most severe deposits. All the deposits that were analysed show a mixed composition. A lot of them came from wells that have recorded ESP intake pressure below or close to the bubble point.

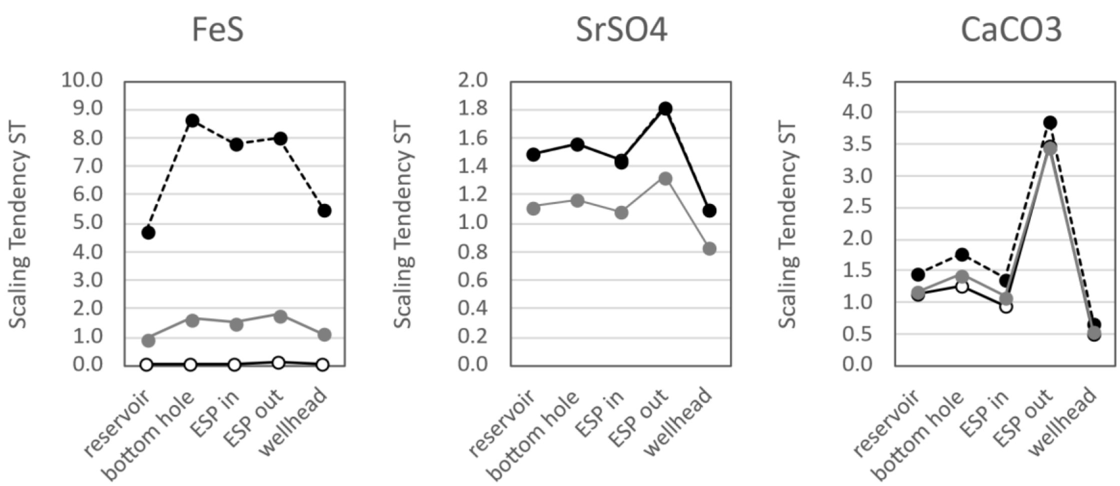

Figure 2—Results of inorganic scale simulations. The scale risk is indicated by the Scaling Tendency (ST) from the reservoir to the wellbore, through the ESP pump to the surface. The simulations included self-scaling with formation water (o), self-scaling with Fe-enriched formation water (•), and breakthrough of injection water (•).

Formation of iron sulfide is not expected with formation water. The brine will become oversaturated with FeS when injection water breakthrough in the oil production wells – but it still below the critical value of ST 3 and not expected to be problematic. With a higher fraction of injection water in the overall water cut this risk will further decrease. However, if the formation water would have a higher Fe^{2+} concentration of 10 mg/l there is a high risk for FeS precipitation from reservoir to the surface. The formation water is oversaturated with $SrSO_4$. The presence of iron is not affecting the risk. Breakthrough of injection water will lower the ST. Although the brine is oversaturated with respect to $SrSO_4$ the ST values are estimated to be below the critical values and calculated amounts of scale are small. In the recent paper of Nichols et al. (2025) a new kinetic model was presented that was validated by laboratory and field data. This paper shows that brines can remain 'metastable' and be oversaturated to a high degree before sulfate scale starts to form. How much oversaturation before precipitation occurs depends on temperature, turbulence and shear stress. The high skin temperature and turbulent flow inside the ESP could result in $SrSO_4$ formation. The risk for sulfate scale in other areas of the well is considered low. However, several of the field deposits contained substantial quantities of acid-insoluble scale. Although not further characterized, it is possible that the acid-insoluble part is sulfate scale. There is a considerable risk for $CaCO_3$ scale formation bottomhole and in the ESP, with formation brine as well as with breakthrough of injection water. The presence of iron has little effect on the scale risk.

The simulations indicate that the Scaling Tendency (ST) does not vary much with water cut. That appears to be aligned with field observations. The water cut ranges from 1.1% in well 23 to >70% in well 2. A scale deposit retrieved from well 23 with only 1.1% water cut shows the presence of inorganic scale, of which one third is acid insoluble. Therefore, the modeled scale risks presented here for an overall water cut of 50% are also representative for lower water cuts as well as for higher water cuts.

The part of the deposit that is either soluble in organic solvents or was combusted by LOI@500C is considered to be (predominantly) hydrocarbons. However, the lab analysis of the field deposits provide no further details on the organic fraction of the deposits. Since new bottomhole samples were collected from a newly drilled well a flow assurance study was done to evaluate the presence of the Asphaltene Onset Pressure (AOP) and Wax Appearance Temperature (WAT).

The WAT was conducted by isobaric cooling runs at 8000, 5000 and 500 psi and at ambient pressure (Table 5). Wax crystals were detected by cross-polar microscopy. The highest WAT was detected at 111F, indicating that the deposits in the well are unlikely to be composed of paraffins as the temperature downhole and tubular are higher. Deposits formed in surface facilities can contain traces of paraffins.

To check if the organic fraction is composed of asphaltenes a SARA analysis was done in wellhead samples, followed by AOP measurements in pressured reservoir fluids. The results of the SARA analysis, presented in table 4, were used to plot the SARA stability screen (Stankiewicz *et al.*, 2002) and calculate the Colloidal Instability Index (Asomaning, 2003). Both screens indicate that the fluid is unstable with respect to asphaltenes. However, two isothermal depressurization tests, at T_{RES} of 170F and 140F showed no asphaltene precipitation. Additionally, no AOP was detected during the isobaric cooling tests. It is therefore concluded that the asphaltenes in this fluid are stable in the well and that the organic fraction of the field deposits is not comprised of asphaltene particles. The organic fraction is most likely just occluded oil, trapped in the inorganic fraction.

Eleven of the 28 wells were not equipped with a Y-tool. The absence of the y-tool limits chemical treatment by bullheading the solvents through the ESP string, or CT after removing the ESP string from the well. For wells equipped with a Y-tool chemical treatment by CT is typically preferred, due to the higher accuracy of placement. For several wells there was limited information on the formation properties (e.g. permeability, porosity). This induces the risk that chemicals might invade high-permeability sections of the open hole and not treat the near wellbore skin area.

The ‘method’ column in the table refers to the chemical treatment method as explained in table 7.

With the knowledge that HAN is not compatible with ESP internals a different BH strategy was developed for naphtha cleaning. To minimize the contact of organic solvents with the ESP the naphtha was bullheaded through the ESP into the open hole, followed by one tubing volume of elastomer-compatible fluid to push the naphtha through and ensure that the ESP is only exposed to compatible fluids.

The ESP intake pressure and well production rate post-cleaning were compared to the values from before the cleaning, and expressed as the percentage gain. A positive value indicates that the ESP intake pressure or production rate is higher than before the cleaning. A negative value indicates that the values are lower than before the cleaning. This can mean that the chemical job was not successful or even caused more impairment.

On average, the cleaning jobs executed in 2 steps were more successful than the single-step jobs. Out of 35 cleaning jobs 19 were executed with a single solvent and 16 with two steps (solvent plus acid). Table 11 below summarizes the results. From the 19 single-step jobs one third had a negative gain compared to the production rate prior to the cleaning. However, only 19% of the dual-step jobs resulted in a negative gain. The 81% success rate of dual-step jobs was considerably higher than the 68% of single-step jobs.

Table 10—Summary table of the chemical treatment results. The P gain % represents the gain in ESP intake pressure, relative to the period prior to cleaning. The Q gain % represents the gain in production rate, relative to the period prior to cleaning. The value is highlighted in grey if the value was also above the original value from the time of ESP commissioning. For some wells the cells are left blank, the original pressure and rates were not available. Method 1, 3, 5, and 6 are single-step cleaning jobs, the others are two-step cleaning jobs.

Well	Date-i	WC2	P gain %	Q gain %	Y-tool	Method	Placement
17	4-Aug-24	25.0	180%	1150%	N	3	BH
1	1-Nov-24	24.4	32%	705%	N	3	BH
17	18-Jan-25	31.6	-59%	679%	N	7	BH
22	10-May-25	16.7	42%	335%	N	7	BH
18	3-Nov-24	23.0	97%	306%	Y	7	BH
24	22-Mar-24	3.5	0%	264%	Y	4	CT
10	26-May-24	41.8		235%	N	1	BH
3	5-Apr-24	61.5		196%	Y	2	CT
4	26-Mar-24	29.8		180%	Y	6	CT
5	21-May-25	67.9	125%	152%	N	7	BH
27	12-Jun-24	21.4	9%	102%	Y	2	CT
20	5-Jul-24	1.5	-18%	92%	Y	4	CT
23	7-Sep-24	1.1	6%	82%	Y	6	CT
17	1-Nov-23	26.7	84%	60%	N	1	BH
21	6-Jan-24	3.4	31%	57%	Y	5	CT
12	10-Jun-25	34.7	6%	51%	N	7	BH
13	4-Apr-25	46.1	53%	48%	Y	7	CT
11	7-May-24	22.4	-13%	43%	Y	7	BH
25	12-Dec-23	37.5	104%	43%	Y	6	CT
14	28-Jun-24	20.8		31%	Y	6	CT
16	2-Apr-25	22.7	29%	25%	Y	7	CT
19	7-Jun-24	12.5	0%	19%	Y	6	CT
9	3-Nov-24	34.5	0%	19%	N	3	BH
28	13-Mar-25	22.7	0%	18%	Y	7	CT
24	01-Sep-24	2.6	4%	9%	Y	6	CT
6	6-Jan-25	56.5		9%	Y	6	BH
26	2-May-25	40.4	6%	-4%	Y	7	CT
8	7-Sep-24	23.8	5%	-6%	Y	6	BH
2	8-Jun-25	70.7	31%	-6%	N	7	BH
7	2-Mar-24	33.1	-13%	-23%	N	3	BH
15	20-Mar-22	27.7	-4%	-25%	N	1	BH
12	2-Jul-23	35.9	-2%	-29%	N	3	BH
13	21-Dec-23	23.1	17%	-30%	Y	3	CT
14	19-Apr-25	40.9	86%	-56%	Y	7	CT
18	25-Mar-24	18.4	-35%	-75%	N	1	BH

Table 11—Comparison of different treatment strategies, with the failure rate for each method. The top two rows show a comparison of single-step (solvent) versus dual-step (solvent & acid) approach. The bottom two rows compare placement by CT versus BH.

	Successful	unsuccessful	failure rate
Single	13	6	32%
Dual	13	3	19%
CT	13	4	24%
BH	13	5	28%

The post-treatment ESP intake pressure and production rates did not correlate to water cut (figure 3). The success rate of cleaning jobs appears to be similar for all wells irrespective of the water cut.

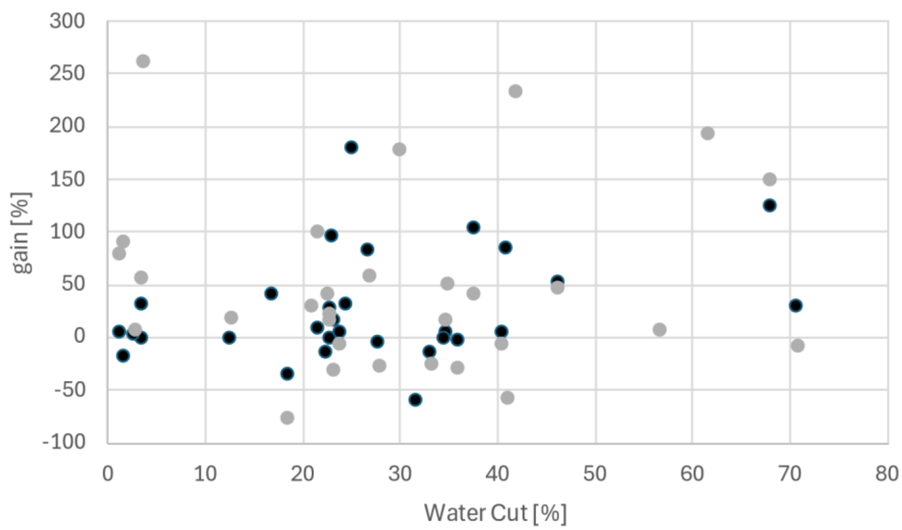

Figure 3—Chemical treatment results as function of water cut. The solid black dots (●) represent the percentage gain in ESP intake pressure P. The grey dots (●) represent the percentage gain in production rate Q.

There was not a big difference in the ESP intake pressure restoration between the single-step or dual-step approach. Both approaches resulted in ~25% increase in intake pressure, compared to pre-cleaning. In 10 out of the 35 jobs the pressure was even restored to above the original intake pressure from the time of ESP commissioning. There was a difference, however, in restoration of the production rates. The single step approach resulted in an average production increase of 70%, compared to an average production increase of 140% for the dual-step approach. (The results of well 17 were excluded from this comparison as the well was almost dead prior to the cleaning job, which skews the results.)

The chemical placement was executed by CT in wells with a Y-tool present. In wells without the Y-tool the chemicals were placed by bullheading through the ESP string. Both placement strategies yielded good results, although the success rate (determined by a net gain in production rate compared to the period pre-treatment) by CT was slightly better (76% versus 72% for BH). The average production gain post-treatment was very similar for both placement strategies (125 to 140% gain).

Figure 4 summarizes the results per cleaning methods. Methods 1, 3, 5 and 6 were a single-step cleaning job with only a solvent or naphtha used for cleaning. See table 7 for more details on the exact chemical mixtures. Methods 1, 5 and 6 yielded similar results of an approximate 50% production gain, implying that applying different types of solvents to attack the organic phase do not make a large difference. Method 3 resulted in a higher production gain, although 3 out of 6 treatments were not successful and the results are heavily dominated by 1 successful treatment. The results from well 17 were excluded, as the 1150% gain is skewed because post-treatment production rates were compared to a pre-treatment rate of almost zero. For all methods using a single-step approach there were cases where the well did not recover to previous production rates. For the dual-step approach the number of successful cases was much higher. Methods 2 and 4 had a 100% success rate. Method 7 was successful in 9 out of 12 wells.

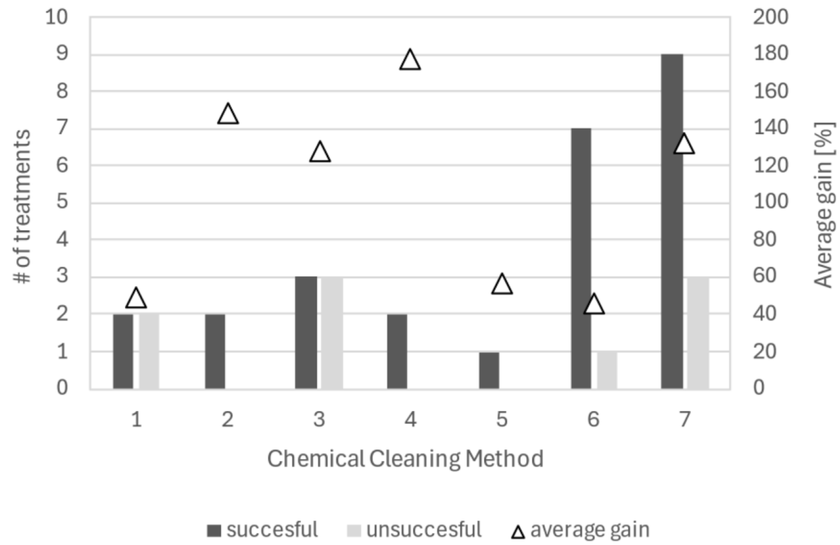

Figure 4—Results of the chemical cleaning, separated by different methods. A summary of the different methods is provided in Table 7. The bars show the number of successful and unsuccessful treatments for each method. The triangles show the average gain in production rate for that method, compared to the rate pre-cleaning. Methods 1, 3, 5 and 6 were single-step, solvent only. Methods 2, 4 and 7 are the dual-step cleaning jobs. Results from well 17 were excluded.

The average production gain for each cleaning method, determined by the production rate post-treatment relative to the pre-cleaning rates, is also better for the multi-step approach. Although method 3 shows an average positive gain, this value needs to be considered with caution. Only 3 out of 6 treatments were successful, and the net gain from one well was very inflated because the post-treatment production rate was compared to the pre-cleaning rate of almost 0 BBL/d. If that outlier is excluded from the comparison the post-cleaning production gain for wells treated with solvent plus acid is double that of wells treated with just solvent.

Conclusions

A series of 28 ESP wells with severe production impairment were used for a chemical treatment pilot. The wells suffer from re-occurring reduction of ESP intake pressure and/or production losses.

To design a chemical treatment program ten field deposits, obtained from wells and multi-phase flow meters, were analysed for the breakdown of the deposit constituents. All deposits are mixed schmoo deposits, with varying concentrations of organic and inorganic constituents. The organic fraction of the deposits is likely to be oil, trapped in the inorganic fraction, as lab tests showed that asphaltene or paraffin particles are not present at these conditions. Both lab tests and scale simulations indicate that the inorganic fraction is a combination of calcium carbonate scale, strontium sulfate scale and iron sulfide.

A chemical treatment program was developed based on several input factors: (a) chemical composition of the deposits, (2) compatibility with ESP components and (3) ESP string configuration. A total of seven different chemical treatment strategies were applied to the 28 wells. The cleaning strategies included single-step solvents and two-step solvent plus acid. Different chemicals were applied, including organic solvents, mutual solvents and dispersants. The acid used was 15% hydrochloric acid. The success of the chemical treatment was expressed by percentage gain of ESP intake pressure and production rate, by comparing the values post-treatment to the values prior to treatment.

The key findings were:

- The success rate of cleaning jobs is not corelated to the water cut of the well.
- The three different solvents used to target the organic fraction (aromatic solvent, dispersant and mutual solvent) yielded similar post-treatment production gains.
- Cleaning jobs executed in 2 steps (solvent followed by acid) were more successful than single-step jobs (solvent or dispersant only). The success rate of dual-step jobs was 81%.
- The average production gain of dual-step jobs was 153%, compared to 73% for single-step jobs.
- The average production gain was very similar for chemical treatment jobs executed by CT or by bullheading.

References

Alharith, A., Albassam, S., & Al-Zahrani, T. (2021). *A Novel Approach for Near Wellbore Stimulation and Deposits Removal Utilizing Thermochemical Reaction*. Presented at the SPE Middle East Oil & Gas Show and Conference, Bahrain. SPE-204771-MS.

Asomaning, S. (2003). Test Methods for Determining Asphaltene Stability in Crude Oils. *Petroleum Science and Technology*, **21**:3-4, 581-590

Bohon, W.M., Blumer, D.J., Chan, A.F., & Ly, K.T. (1998). *Novel chemical dispersant for removal of organic/inorganic "schmoo" scale in produced water injection systems*. Presented at Corrosion 98, NACE-98073.

Chen, T., Wang, Q., Chang, F., Leal, J., & Espinosa, M. (2019). *Root Cause Analysis for Iron Sulfide Deposition in Sour Gas Wells*. Presented at the SPE Kuwait Oil & Gas Conference and Show. SPE-198187-MS.

Eroni, V., Anfindsen, H., & Mitchell, F. (2015). *Investigation, Classification and Remediation of Amorphous Deposits in Oilfield Systems*. Presented at the SPE International Symposium on Oilfield Chemistry, Texas. SPE-173719-MS.

Horsup, D.I., Dunstan, T.S, & Clint, J.H. (2007). *A Break-Through Corrosion Inhibitor Technology For Heavily Fouled Systems*. Presented at NACE Corrosion Conference and Expo, Tennessee. Paper 07690.

Hsi, C.D., Dudzik, D.S., Lane, R.H., Buettner, J.W., & Neira, R.D. (1994). *Formation Injectivity Damage Due to Produced Water Reinjection*. Presented at the SPE International Symposium on formation Damage Control. SPE-27395.

Jones, C.R., Downward, B.L., Hernandez, K., Curtis, T. & Smith, F. (2010). Extending performance boundaries with third generation THPS formulations. *NACE International.* Paper No. 10257.

Jordan, M.M. Johnston, C.J., Bennett, B., & Cole, G. (2022). *Removal of Iron Sulphide/biomass Via Continual Injection Dissolver/dispersant, A North Sea Case Study*. Presented at the SPE International Oilfield Scale Conference and Exhibition. SPE-209478-MS.

Littmann, E.S. (1987). *Control of microbiologically influenced corrosion in oilfield production equipment*. Presented at the 62nd Annual Technical Conference and Exhibition of the SPE. SPE-16909.

Lu, H., Wei, W., Wei, W., Leach, D., & Yan, C. (2024). *Overview of FeS scale control and treatment under challenging conditions*. Presented at the SPE Oilfield Scale Symposium, Scotland. SPE-218728-MS.

Nichols, D.A., Fyfe, A.A., Stiven, S.; Jordan, M.M. (2025). *A Field Case Study Validating the Use of a Kinetic Scaling Risk Model to Predict the Metastability of Barium Sulfate*. Presented at SPE International Conference on Oilfield Chemistry. SPE-224255.

Reeves, N.K. (1989). *A study of bacteria-influenced scale deposition*. Presented at the SPE Joint Rocky Mountains Regional/Low Permeability Reservoirs Symposium and Exhibition. SPE-18989.

Stankiewicz, A.B., Flannery, M.D., Fuex, N.Q., Broze, G., Couch, J.L., Dubey, S.T., Leitko, A.D., Nimmons, J.F., Iyer, S.D., Ratulowski, J., & Westeric, J.T. (2002). *Prediction of asphaltene deposition risk in E&P operations*. Presented at the AIChE Spring National Meeting: 3rd International symposium on mechanisms and mitigation of fouling in petroleum and natural gas production, pp. 410-416.